Where a Quantum Reservoir Works: A Transferable Operating Band

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***Abstract*—In quantum reservoir computing, a fixed quantum system transforms an input signal, while learning reduces to training a simple linear readout on its measured outputs. Since the quantum dynamics themselves are never optimized, the method is well suited to today's hardware. Yet these dynamics must still be chosen carefully, because their settings remain fixed throughout training and inference. It therefore remains an open question where, in its control space, a fixed quantum system supports learning well. We address this question for a dissipative reservoir by mapping performance over three central physical controls: the strength of the input drive, the coupling between neighboring qubits, and the rate of dissipation. Good performance concentrates in a single, well-defined operating region of this control space. This region transfers across tasks and reservoir initializations, and the same regime persists under an architectural change. It is also mechanistically grounded, since it disappears whenever any of the mechanisms that create it is removed. Finally, the region can be located cheaply before any task is run, using a simple memory diagnostic.**

***Index Terms*—quantum machine learning, reservoir computing, quantum computing, dissipative systems, memory capacity**

I. Introduction

Reservoir computing builds on a simple principle: a fixed dynamical system, left untrained, transforms an input signal into a high-dimensional record of its recent history, and learning reduces to fitting a single linear readout on top of that record [1]–[3]. This simplicity makes the approach appealing for near-term quantum hardware: native quantum evolution can serve as the reservoir, and only the classical readout must be trained [4], [5]. At the same time, it places the full burden on the fixed dynamics, raising an open question: what makes those dynamics useful?

Recent QRC work spans noisy gate-model, photonic, and continuous-variable platforms, and protocols from weak measurement to engineered dissipation and optical memory control [5]–[7], [9], [13]. Across this range the question persists, and classical reservoir theory already supplies the warning: maximal expressivity alone does not make a reservoir useful. A reservoir works by holding a balance, retaining recent input, forgetting old input in a controlled way, and exposing nonlinear functions of that history to the readout [3], [10]. A quantum reservoir inherits this lesson but complicates it, because these ingredients are no longer free dials. They are consequences of distinct physical resources: how strongly each input perturbs the carried state, how coherently the system mixes that state across its parts, and how quickly it is allowed to forget [5], [13]. Closest to our question, dynamical-phase analyses of Hamiltonian spin-network reservoirs locate good operation in the thermal phase near the thermalization transition [8]. We pose the complementary question for an engineered, per-step dissipative gate-model reservoir and add transfer, ablation, and screening tests beyond a phase map.

For a stateful dissipative gate-model reservoir, we treat this three-resource space as the proper setting for asking where the reservoir works, and we map it. What emerges is not a single tuned operating point but a connected region, characterized by gentle input, nonmaximal coupling, and controlled damping, whose boundaries stay essentially unchanged from task to task. The contribution of this paper is that region itself: (i) its existence as a single connected band; (ii) its transfer across tasks, reservoir initializations, and a chronological holdout; (iii) its mechanistic grounding, since every single-ingredient ablation destroys it; and (iv) its predictability, since a task-free memory diagnostic locates it before any task-specific search.

II. Protocol

A. Reservoir

The reservoir is a small qubit system run as a recurrent loop. At each step it receives one scalar input value, evolves, and hands its state forward without ever being reset, so the carried state holds a record of recent inputs. We write $\rho_{t,\ell}$ for the state after input t has passed through layer ℓ; with L the final layer, $\rho_{t-1,L}$ is handed forward from the previous step.

At step t, the input u_t is written into this carried state by the same Y rotation on every qubit,

$$\rho_{t,0} = U_{\text{in}}(u_t)\,\rho_{t-1,L}\,U_{\text{in}}^{\dagger}(u_t), \qquad U_{\text{in}}(u_t) = \bigotimes_i R_y(\beta u_t), \tag{1}$$

with β the input drive.

The perturbed state then passes through L identical layers, each a frozen local mixer U_{mix}, a ring ZZ coupling U_λ, and amplitude damping D_γ applied to every qubit,

$$\begin{aligned} \rho_{t,\ell} &= D_\gamma^{\otimes n}\left[U_\lambda U_{\text{mix}}\,\rho_{t,\ell-1}\,U_{\text{mix}}^{\dagger}U_\lambda^{\dagger}\right], \\ U_\lambda &= \prod_{(i,j)\in E_{\text{ring}}} e^{-i\lambda Z_i Z_j}, \end{aligned} \tag{2}$$

with E_{ring} the nearest-neighbor edges of the qubit ring, λ the coupling strength, and γ the damping rate. Everything else is frozen: we vary only $\theta = (\beta, \lambda, \gamma)$, with β and λ angles in radians and $\gamma \in [0, 1]$ the amplitude-damping probability, and train only a linear readout. Each control targets one ingredient reservoir theory requires: β sets how strongly each new input is written in without overwriting what the state already carries, λ generates nonlinear mixing of past inputs, and γ provides the controlled forgetting on which fading memory depends.

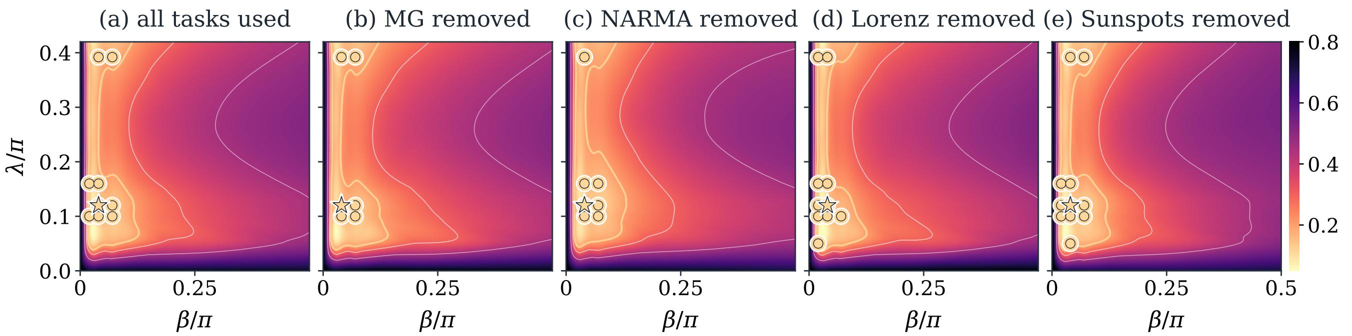

Fig. 1. The good operating region is essentially the same whichever task we score. Each panel maps validation rank over input drive β and coupling λ at fixed damping $\gamma = 0.12$. Brighter means better rank. Panel (a) uses all four tasks; panels (b) to (e) each drop one. Gold circles mark the top validation quintile across tasks and seeds; the star is the band's medoid. The basin and circles barely move when a task is dropped, so the band is not an artifact of any single task.

B. Configuration

Unless stated otherwise, the reservoir has four qubits and two layers with ring coupling, uniform input injection, and amplitude damping; the readout is ridge regression on single-qubit Z and ring-pair ZZ expectations alone. We sweep the three controls β, λ, γ on a fixed grid and select every hyperparameter, including the ridge penalty, on validation data only; the holdout never defines the operating region. Crossing the four benchmark tasks below with twenty frozen-reservoir seeds yields a balanced panel of 80 task-seed replicates per grid coordinate. All grid coordinates and task–seed replicates are fixed before performance is inspected.

C. Tasks, Metric, and Splits

We use four scalar one-step-ahead prediction tasks spanning distinct demands: the Mackey–Glass and Lorenz systems for smooth chaos [14], [15], NARMA for nonlinear memory [16], and annual sunspot counts for irregular real-world data. Inputs are scaled to $[-1, 1]$ and targets standardized; each task is split chronologically into washout, train, validation, and holdout segments. We score with normalized mean-squared error (NMSE), the mean squared error divided by the target variance, so that predicting the mean is the no-skill baseline and lower is better. NMSE is computed separately on validation and holdout.

D. Operating Band

The four tasks produce errors on very different scales, so raw NMSE values cannot be pooled across them; following standard practice for comparisons across data sets [11], we compare settings by rank instead. Within each task j and seed s, a setting $\theta \in \Theta_{\text{grid}}$ receives the validation percentile rank $r_{j,s}(\theta)$, and lower is better. To guard against flukes—one easy task or one lucky seed—we keep every setting that frequently reaches a high rank rather than crowning a single winner. The operating band contains the settings that land in the top fraction p for at least a fraction q of task-seed pairs,

$$B_{p,q} = \{\theta : \Pr_{j,s}[r_{j,s}(\theta) \le p] \ge q\}. \tag{3}$$

We call $B_{p,q}$ the band, and the dynamical behavior it identifies the operating regime. This frequency criterion mirrors stability selection [12] and treats the band as a search prior over settings that reliably work. We report the strict band $B_{20,0.7}$, the settings that reach top-quintile rank on at least 70% of the task-seed replicates. We fixed these thresholds before the analysis, and the band is not sensitive to them: relaxing either one enlarges the same region rather than revealing new ones, from 22 grid coordinates at $B_{20,0.7}$ to 41 at $B_{20,0.6}$ and 70 at $B_{30,0.7}$. In the transfer tests the band is built from the remaining replicates and scored on the withheld one; in the holdout audit it is frozen on validation, then evaluated once on holdout. Because ranks are formed within each task-seed replicate before aggregation, no large-error task or difficult seed can dominate the band.

III. Results

Settings with low validation rank form one compact, connected basin rather than scattered optima, and the basin barely moves when any single task is dropped (Fig. 1). The damping coordinate is selected by validation: the near-zero slice $\gamma = 0.01$ contains almost no band points, while the strict $B_{20,0.7}$ core fills in from $\gamma = 0.12$ to 0.30 with medoid $\gamma = 0.22$ (Fig. 2). Fig. 1 plots $\gamma = 0.12$, the first populated slice; the medoid slice appears in Fig. 2(c).

To test generalization beyond a joint fit, we rebuild the band with part of the data withheld and score it on exactly that part. When one task is withheld, the band built from the rest scores mean held-out rank 0.193, CI $[0.176, 0.210]$; when one seed is withheld, 0.151, CI $[0.139, 0.163]$. With 0.5 the chance level, both values sit deep in the top quartile: the band transfers across task identity and reservoir initialization alike. It also holds on unseen data: frozen on validation and evaluated once on the chronological holdout, it remains in the top quartile on every task (Table I), including NARMA, whose large raw NMSE reflects task difficulty rather than band failure. The regime, though not its coordinates, also survives an architectural change. A four-qubit, three-layer variant forms a band whose held-out performance matches the base model (0.154 vs. 0.153), yet the two bands share almost no grid coordinates (Jaccard overlap 0.04). Coordinate-level transfer thus fails, as expected once

TABLE I
HOLDOUT PERFORMANCE AFTER VALIDATION-ONLY BAND SELECTION. RANK IS NORMALIZED SO THAT 0.5 IS CHANCE AND LOWER IS BETTER.

Task	Band NMSE	Medoid NMSE	Band rank
Mackey–Glass	7.05e-05	5.26e-05	0.104
Lorenz	1.51e-05	1.52e-05	0.136
NARMA	0.560	0.607	0.204
Sunspots	0.209	0.209	0.168
All	0.192	0.204	0.153

an added layer changes the per-step dynamics; what carries over is the regime, and the band need only be relocated for each architecture, which the diagnostic below makes cheap. To test whether the band reflects the reservoir mechanisms rather than a coincidental grid location, we repeat the selection after ablating one ingredient at a time: no damping, no recurrent mixer, dephasing or depolarizing noise in place of amplitude damping, or Z-only/ZZ-only readouts. Figure 3 shows three representative cases (no damping, no mixer, dephasing); the rest run under the same criterion but are not plotted. In the base model, the plotted $\gamma = 0.12$ slice contains nine band coordinates with mean rank 0.223, well inside the top quartile. Under every ablation, however, the re-selected band is empty: no coordinate is chosen often enough to satisfy the same $B_{20,0.7}$ stability criterion.

The original band coordinates also lose their advantage once the mechanisms are removed. Their mean rank deteriorates to 0.473 without damping, 0.445 with dephasing, 0.509 with depolarizing noise, 0.367 without the recurrent mixer, 0.486 with Z-only readout features, and 0.468 with ZZ-only features. Even the mildest case falls outside the top quartile, and most sit near chance. This rank loss concentrates exactly where the base band sat (Fig. 3): the ablations break precisely the region that was useful. Together these results indicate the regime emerges from the joint action of write-without-overwrite input, nonlinear mixing, controlled forgetting, and a readout accessing both local and pairwise observables.

Finally, the regime can be located before any task is run. Memory capacity is the summed held-out R^2 for reconstructing delayed inputs from the readout features; information-processing capacity extends it to polynomial functions of the history [10]. The task-free memory map reproduces the band from the dynamics alone (Fig. 4a); across all settings and seeds, memory capacity strongly predicts validation rank ($\rho_s = -0.88$), as do information-processing capacities, while raw feature diversity does not (Fig. 4b,c). A cheap memory measurement therefore screens for good settings before any task-specific search.

IV. DISCUSSION

For this reservoir family, where the reservoir works is set by a transferable operating regime, not by task-specific tuning: the same region holds across all four benchmark tasks, and it persists when any one task or seed is withheld. The regime is picked out by memory; raw feature diversity, an expressivity proxy, does not predict it. What matters is the balance classical reservoir theory identified, with drive, coupling, and dissipation holding memory and nonlinearity in tension [3], [10]. The same balance explains the rest: strip out any single mechanism and the regime collapses.

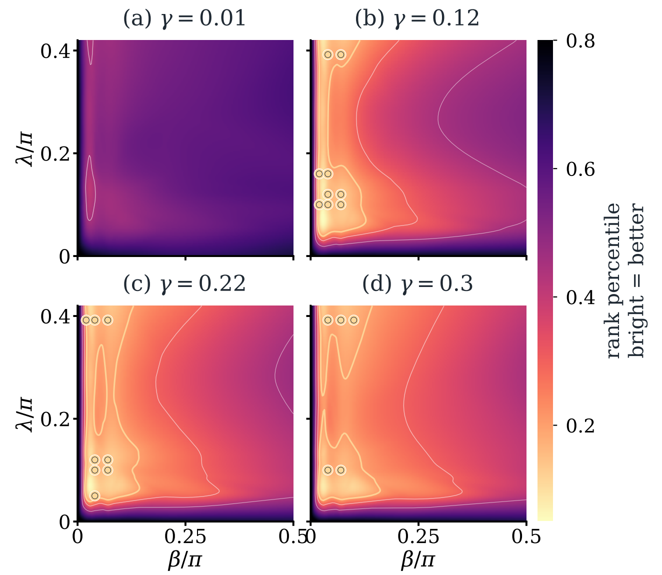

Fig. 2. The band requires nonzero damping. Each panel maps validation rank over (β, λ) at the damping rate γ named in its title (brighter = better); gold circles are the $B_{20,0.7}$ band points at that γ. The band is essentially empty at $\gamma = 0.01$ and fills in from $\gamma = 0.12$ to 0.30.

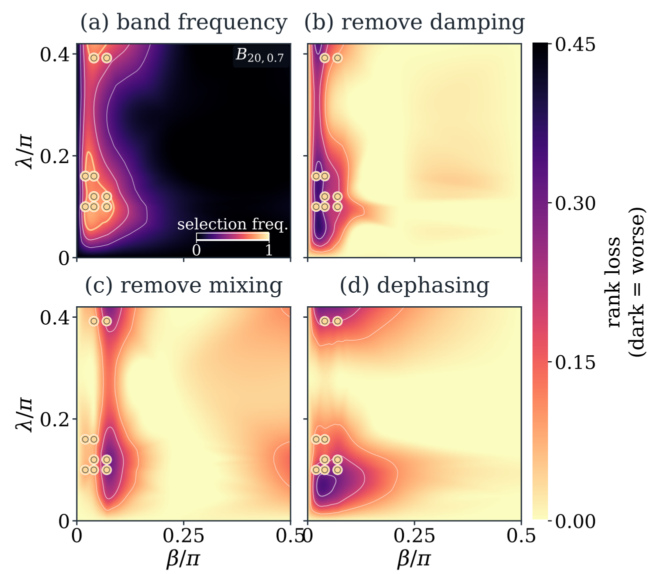

Fig. 3. Mechanism ablations destroy the operating band. (a) Selection frequency in the original reservoir: for each coordinate (β, λ), the color shows how often it reaches top-quintile validation rank across the 80 task–seed replicates; gold circles mark the $B_{20,0.7}$ band. (b)–(d) Validation-rank loss for three representative ablations named in the panels, with darker colors indicating larger deterioration. The strongest losses occur at the original band coordinates.

Our claims are restricted to simulated instances of this architecture family; finite-shot effects, calibration drift, device noise, and readout cost are not addressed here [7]. Because the readout features are low-weight Pauli expectations, shot noise enters at the standard $1/\sqrt{S}$ rate, and whether the

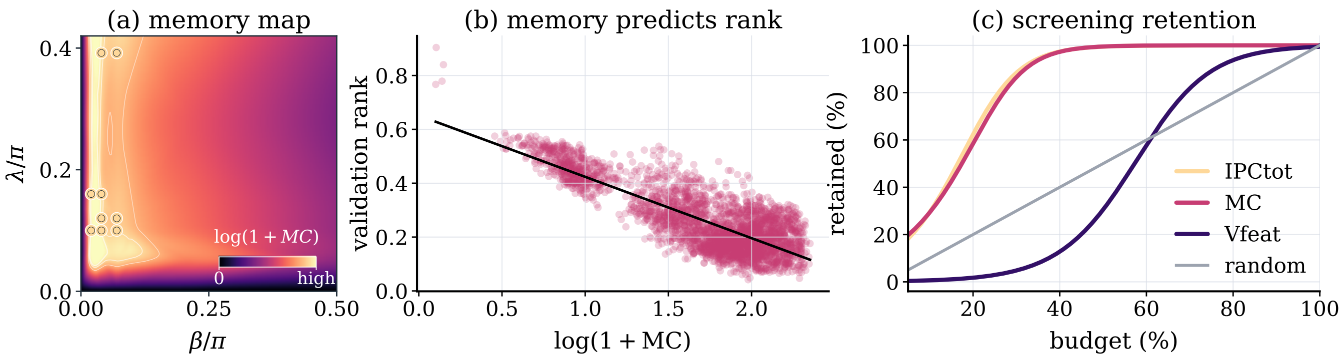

Fig. 4. Memory alone locates the band before any task is run. (a) Task-free memory-capacity map, $\log(1+\mathrm{MC})$, on the $\gamma = 0.12$ slice of Fig. 1; gold points lie in the memory-rich region. (b) Higher memory predicts lower validation rank (Spearman $\rho_s = -0.88$). (c) Screening by each diagnostic's top-ranked fraction: MC and information-processing capacity (IPC) recover good settings far faster than random selection, whereas raw feature diversity (V_{feat}) barely beats chance.

band survives realistic shot budgets is the natural next test. At this size the reservoir is also classically simulable, so the contribution is methodological, not a claim of quantum advantage. Within these limits, the practical consequence is concrete. The default workflow in QRC, a fresh hyperparameter search for every task, treats each problem as if the reservoir had to be rediscovered from scratch; our results say it does not: one memory measurement, taken once and without any task data, identifies the settings that then serve every task we tested. How useful a dissipative quantum reservoir is depends far more on where it is operated than on the fine details of its architecture, and where to operate it can be read off its memory alone.

Reproducibility Appendix

A. Package

The anonymous reproducibility repository contains everything needed to reproduce every number and figure: the simulator, selection and diagnostic code, fixed grids and seeds, checked-in CSV/JSON results, and figure scripts, with all stochastic components seeded for deterministic reruns.

B. Simulator and diagnostics

We simulate the full density matrix exactly, without finite-shot sampling. The state starts from $\rho_0 = |0\cdots0\rangle\langle0\cdots0|$, is carried between input steps without reset, and each seed uses one fixed local mixer $U_{\mathrm{mix}} = \bigotimes_i R_x(\theta_i)$ with $\theta_i \sim \mathcal{U}[0, 2\pi)$, reused at every step and layer. The readout uses exact Z_i and nearest-neighbor ring Z_iZ_j expectations plus an unpenalized intercept. All model selection, including ridge penalties and band construction, uses validation NMSE only; holdout is evaluated only after the operating band is frozen. The strict band $B_{20,0.7}$ contains 22 coordinates in the full three-dimensional (β, λ, γ) grid, while the mechanism-ablation maps in Fig. 3 use the plotted $\gamma = 0.12$ slice containing 9 of them. Task-free memory and information-processing diagnostics are computed on an independent random drive with chronological splits; negative held-out R^2 values are clipped to zero.

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